

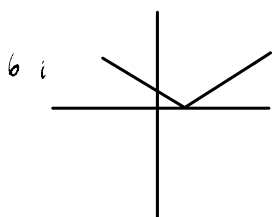
Algebra 6

Consolidation

4. i $e^{2x} = g \circ f(x)$
 ii $e^x = f \circ g(x)$
 iii $e^{2x} = g \circ f(x)$
 iv $e^{4x} = f \circ g(x)$

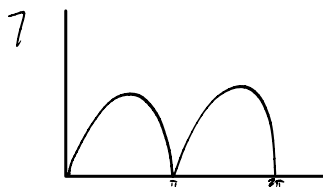
5 $f(x) = 2x$
 $g(x) = \sinh x$

i $f \circ g(x) = 2 \sinh x$
 ii $(f \circ g)^{-1}(x) = \operatorname{arcsinh}(\frac{x}{2})$



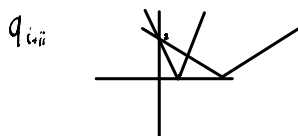
ii $3 - \frac{1}{2}x > 1$ or $3 = 1 - \frac{1}{2}x$
 $x = 8$ or $x = -4$

iii $|x-2| \leq 6$



ii) 4

8 $|3x-2| > 1$
 $x > 1$ or $x < \frac{1}{3}$



iii $|6x-2| < |x-5|$
 $9x+2 < 0: x < -\frac{2}{9}$
 or $7x-7 < 0: x < 1$

Exam Qs

1
 c) $f(-1) = 8$
 $f(8) = \ln 40$

d) The mean function would be one-to-many, which is disallowed.

2 $f(x) = x^2 + 8ax + 4a^2$
 $g(x) = 6x - 2a$

a) $f \circ g(x) = (6x-2a)^2 + 8a(6x-2a) + 4a^2$
 $= 36x^2 - 24ax + 4a^2 + 48ax - 16a^2 + 4a^2$
 $= 36x^2 + 24ax - 8a^2$

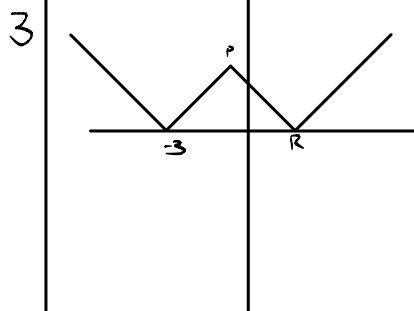
range $\rightarrow 6^2 - 4ac \geq 0$
 $(24a)^2 - 4 \times 36 \times 8a^2 \geq 0$
 $576a^2 - 1152a^2 \geq 0$
 $-576a^2 \geq 0$
 $a \geq 0$

6) $36(x)^2 + 24(x)a - 8a^2 = 144$

$8a^2 - 48a = 0$
 $a(8a - 48) = 0$
 $a = 0$ or $a = 6$

a is surely positive, $a = 6$?

c) No inverse: functions can't be one-to-many



c) $2 - |x+1| = 0 \rightarrow |x+1| = 2; x = -3$ or $x = 1$
 $\therefore R = (-3, 0) \quad Q = (1, 0)$

$P = 2 - \text{min value of } |x+1|$
 $= 2 - 0 = 2 \quad (\text{when } x = -1, |x+1| = 0)$

d) $2 - |x+1| = \frac{1}{2}x$

$|x+1| = 2 - \frac{1}{2}x$
 $x^2 + 2x + 1 = \frac{1}{4}x - 2x + 4$ square both sides
 $\frac{3}{4}x^2 + 4x - 3 = 0$
 $3x^2 + 16x - 12 = 0$
 $(3x-2)(x+6) = 0$
 $x = \frac{2}{3}$ or $x = -6$